Weekly Report

Review of GEMini Clustering Code

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TRIUMF DarkLight Group Meeting

July 7, 2026

Summary

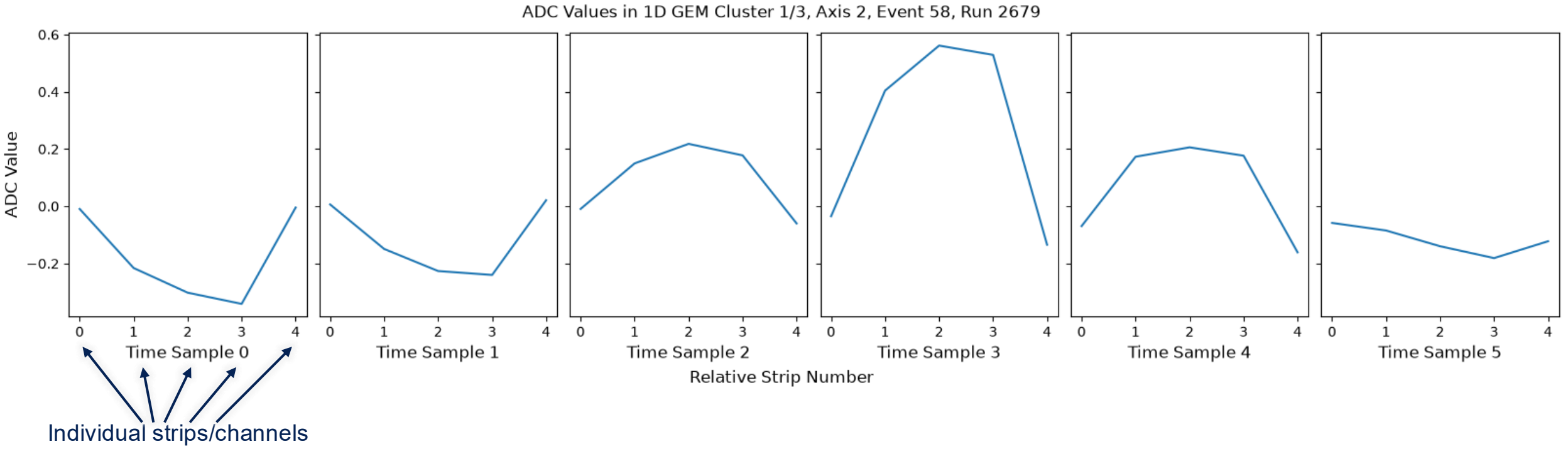
1. Recap of last week
2. How does our cluster identification algorithm work?
3. Next steps to improve GEMini

Review of last week

How the channel values in a cluster change over time

Pedestal-subtracted values across five strips over six time samples

These five strips were identified as containing a cluster in this event

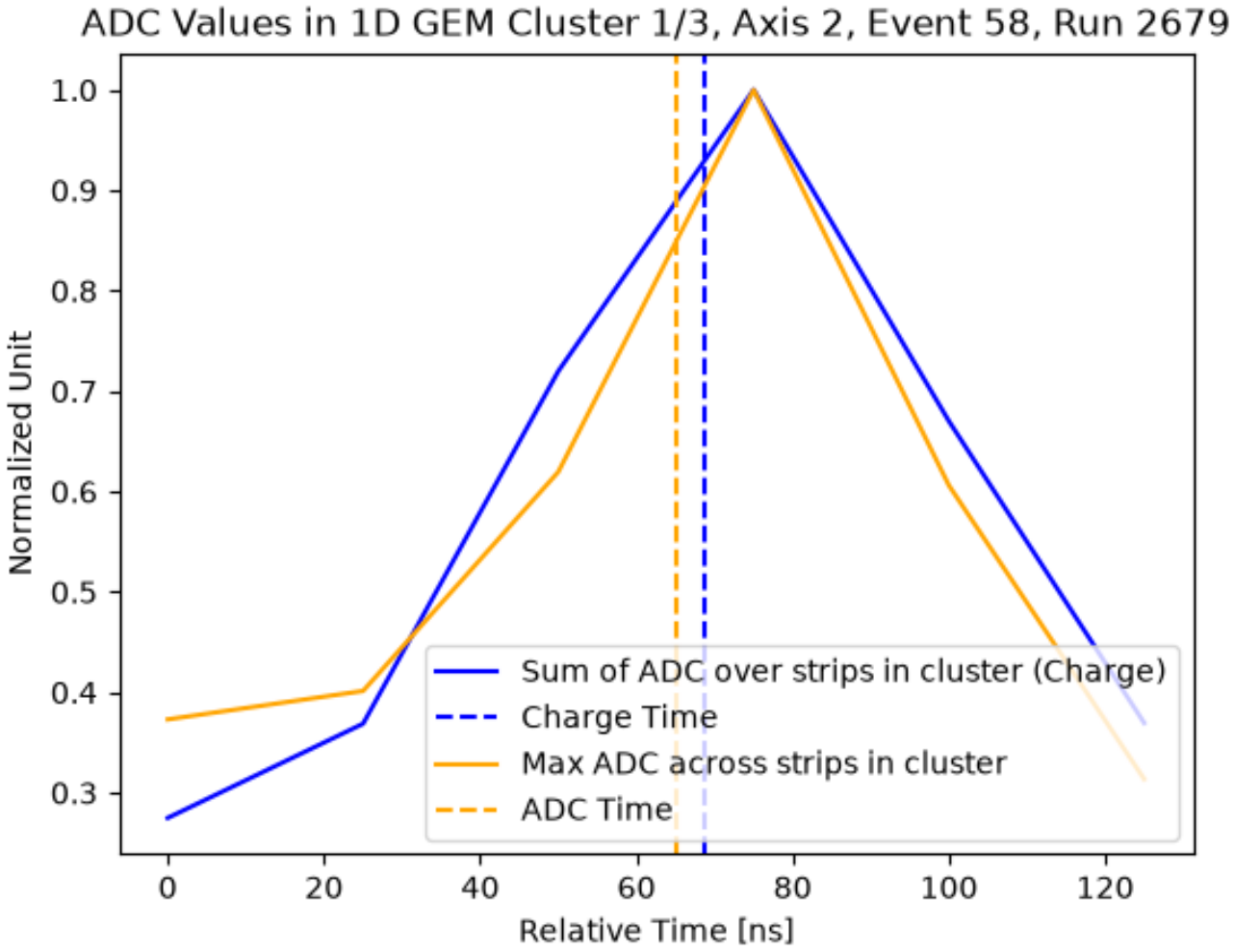


Time

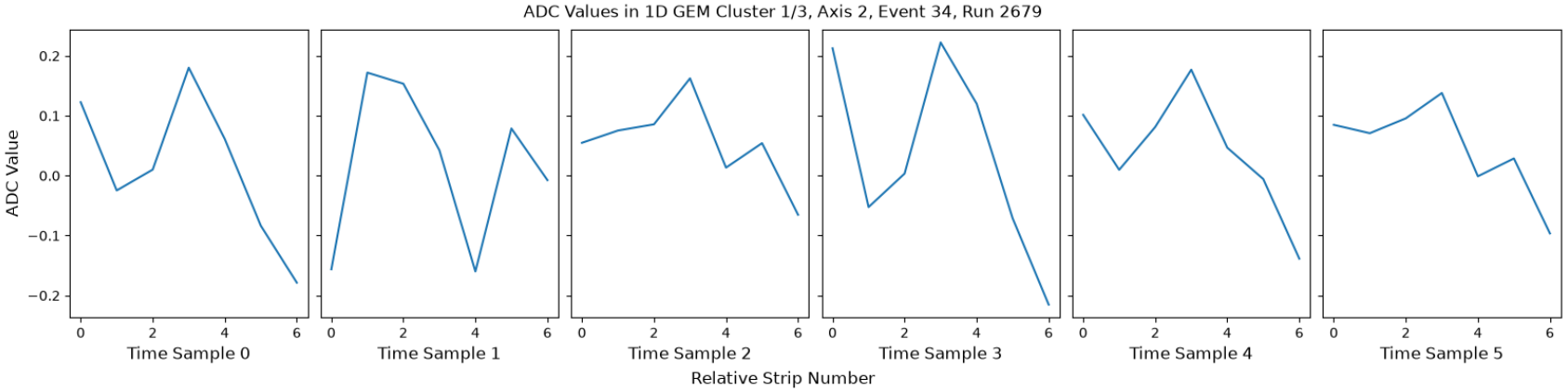
Approximating GEM cluster time

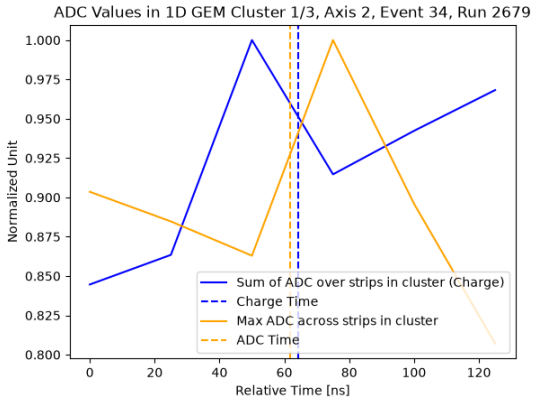
“Charge Time” – time weighted by sum of channel values for each sample

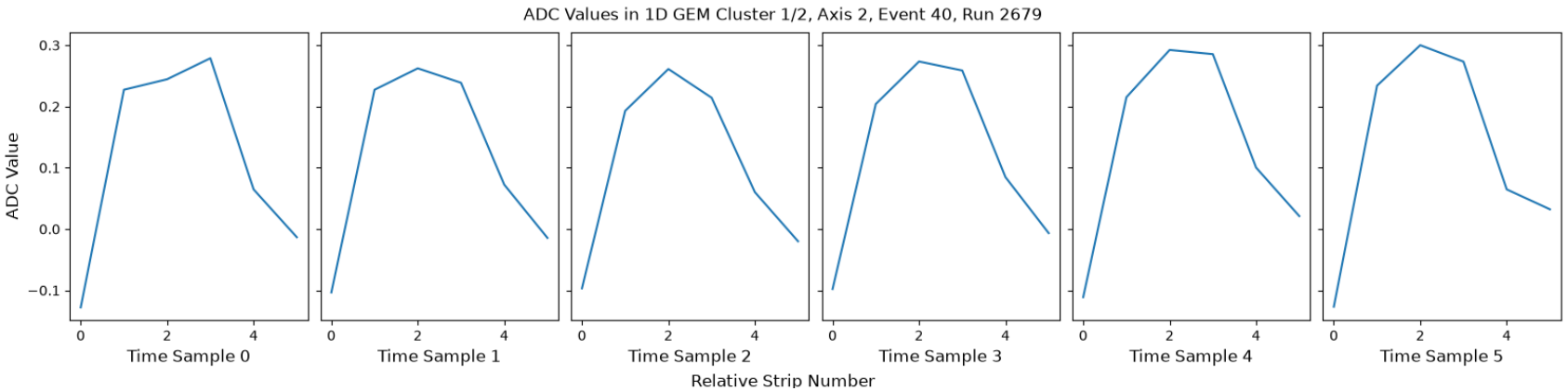
“ADC Time” – time weighted by maximum channel value in each sample

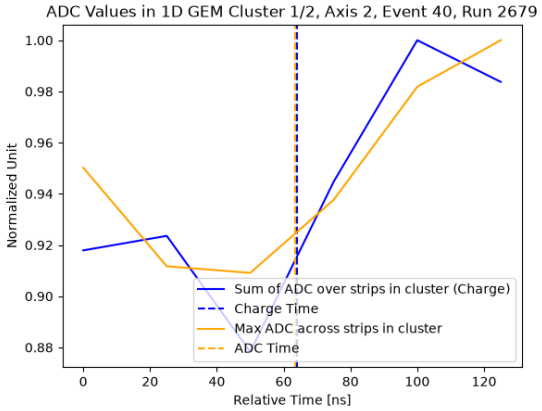


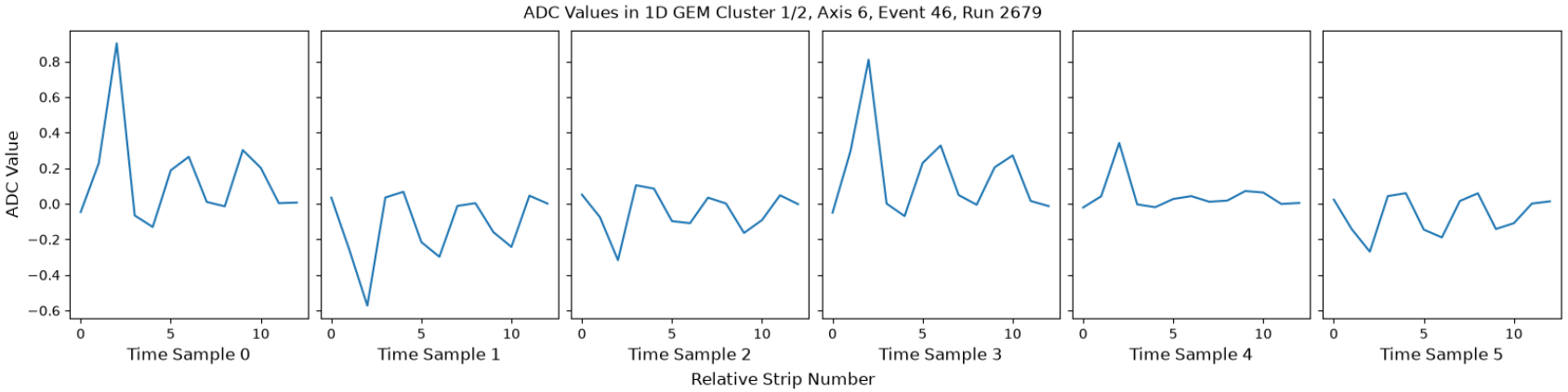
Variety of cluster signal shapes

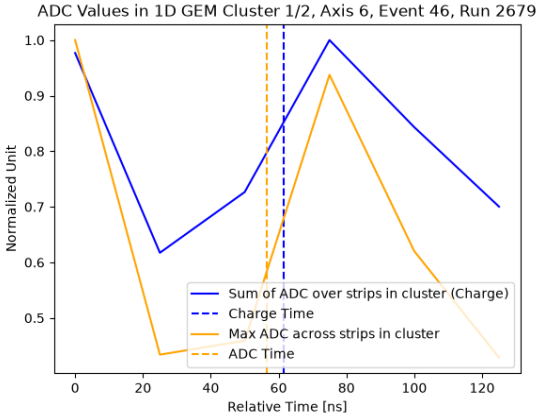










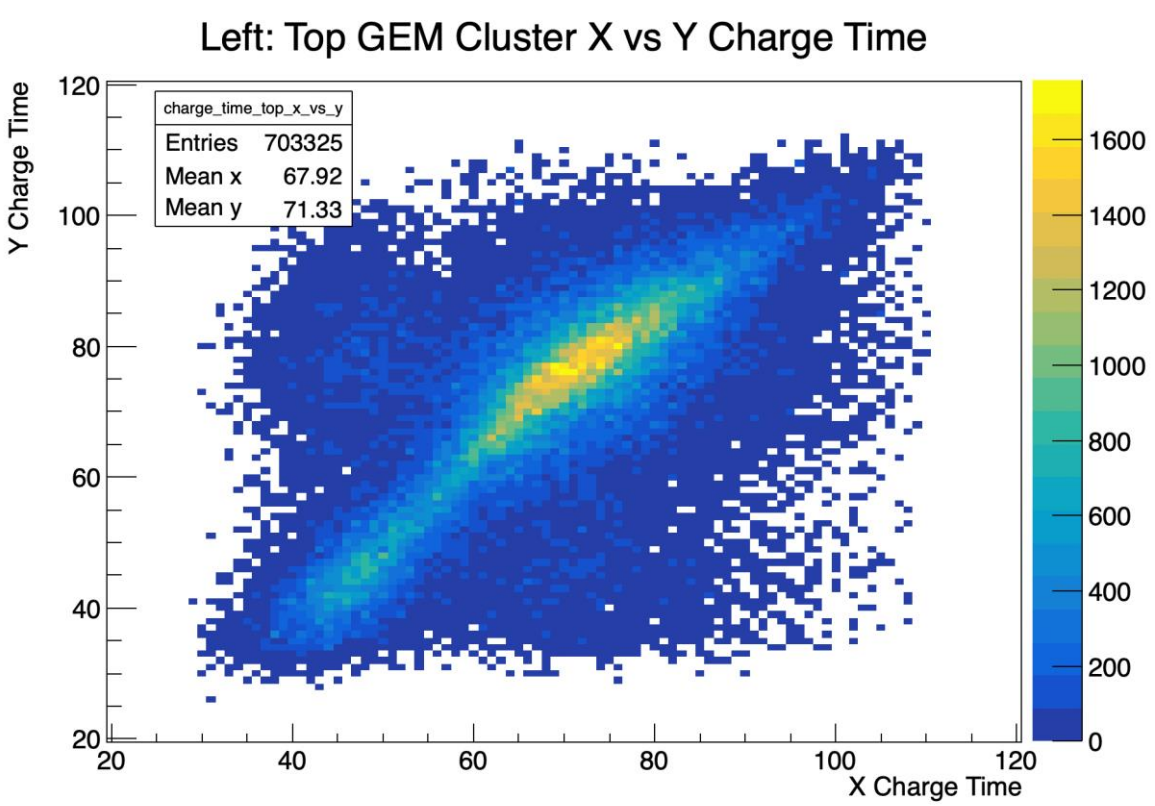


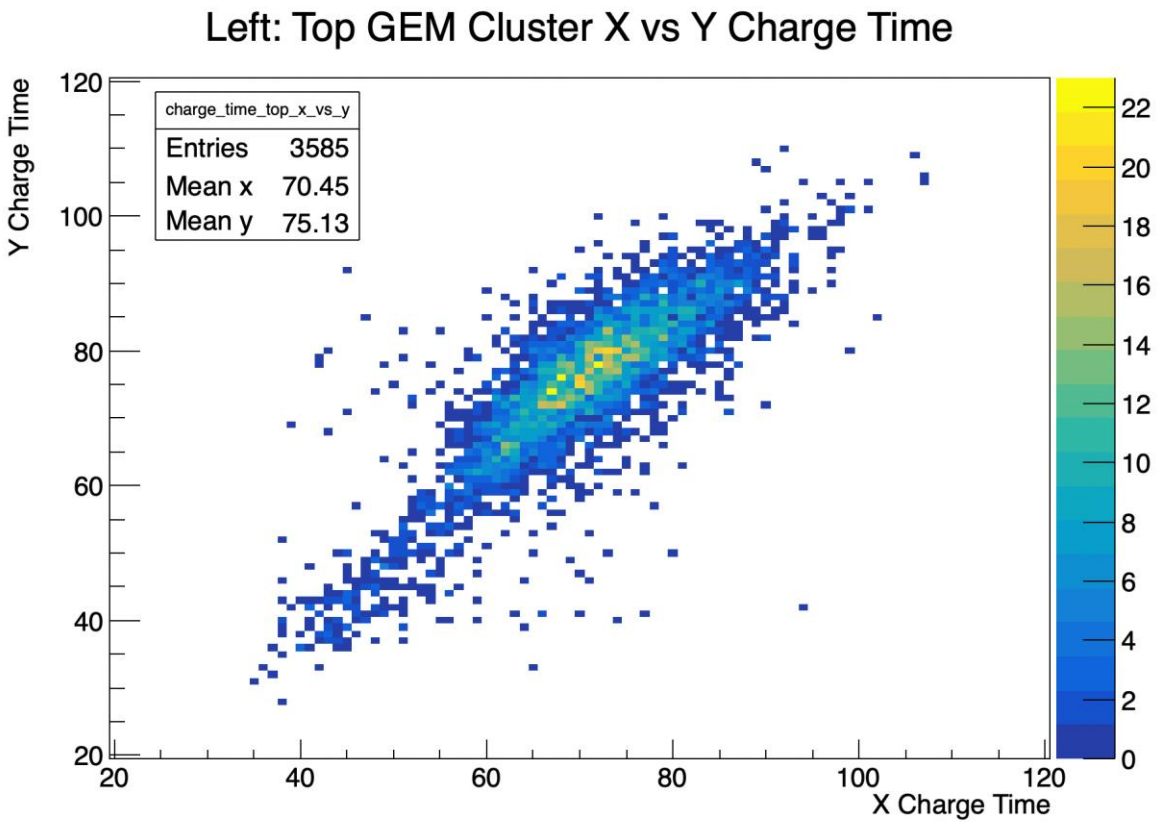
Correlation Between X and Y axis Cluster Times

Charge Time

Run 4010 - Elastic

Left Top GEM clusters





All tracks

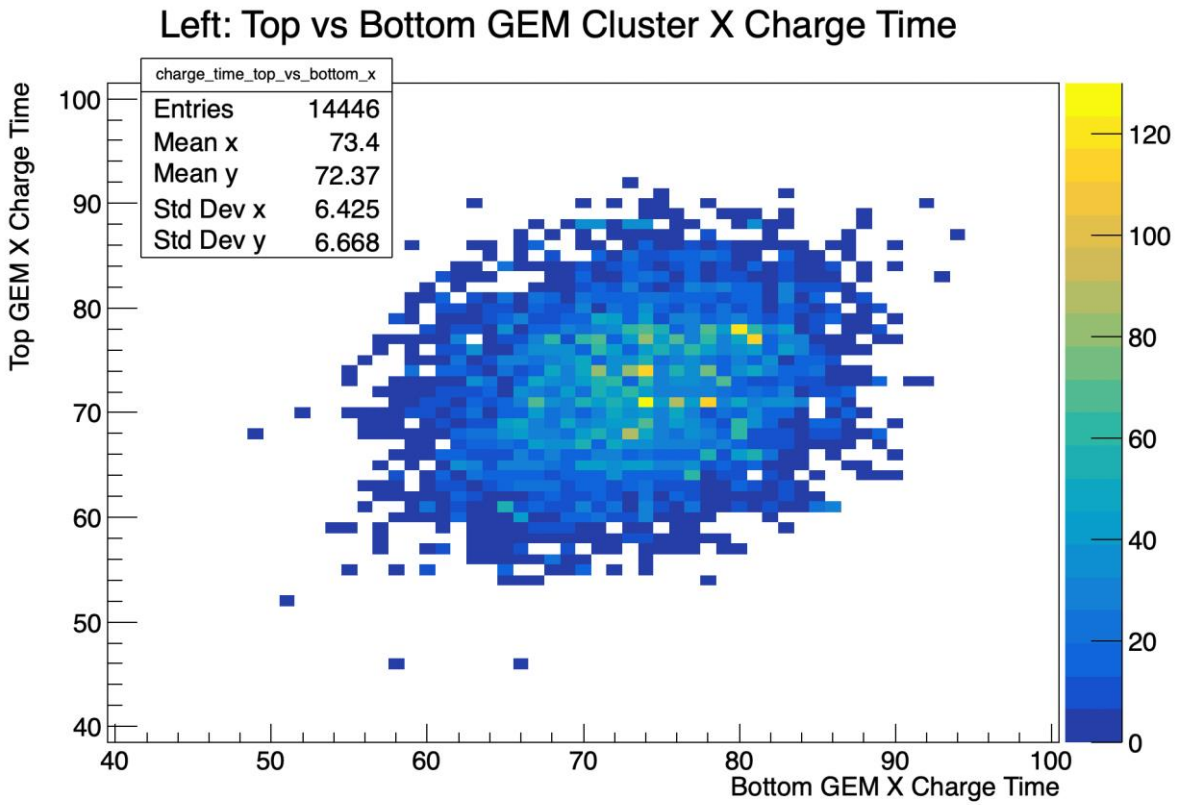
Single (one hit per detector per event) and On-Paddle (hit scintillators) tracks

Correlation Between Bottom and Top GEM Cluster Times

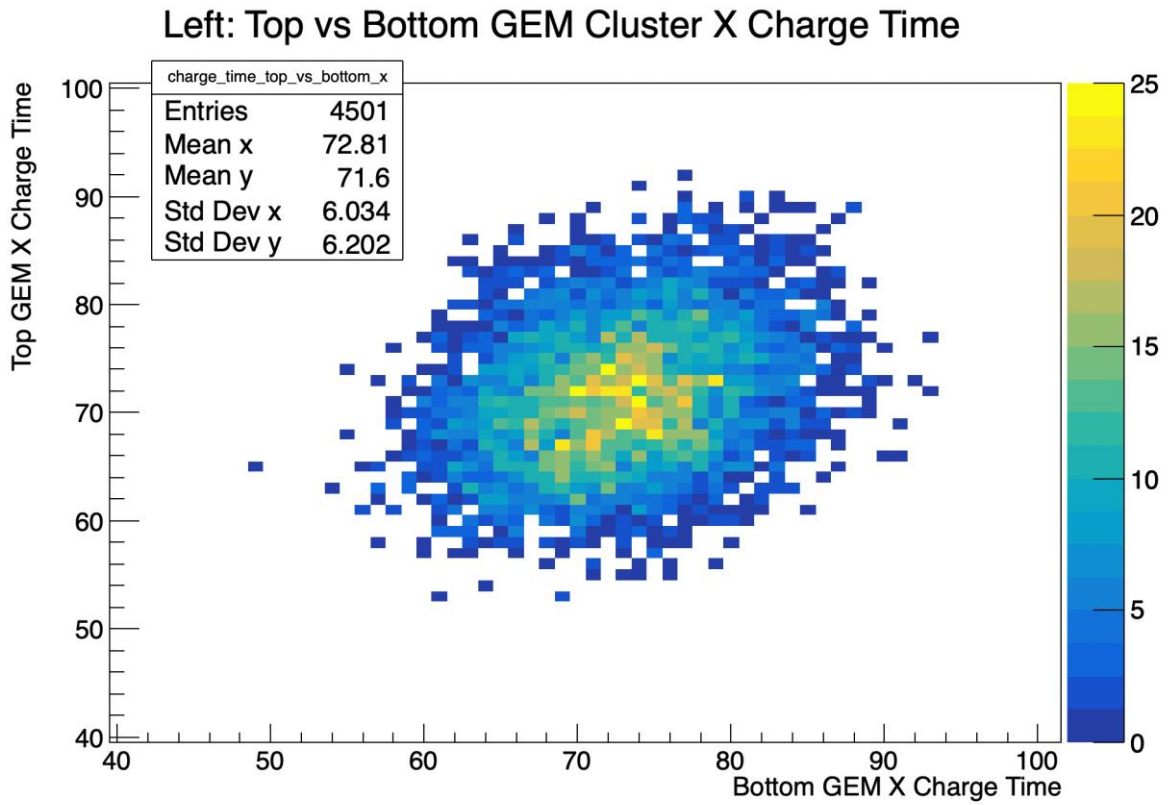
Charge Time

X axis clusters

Run 3622 - Cosmic



All tracks



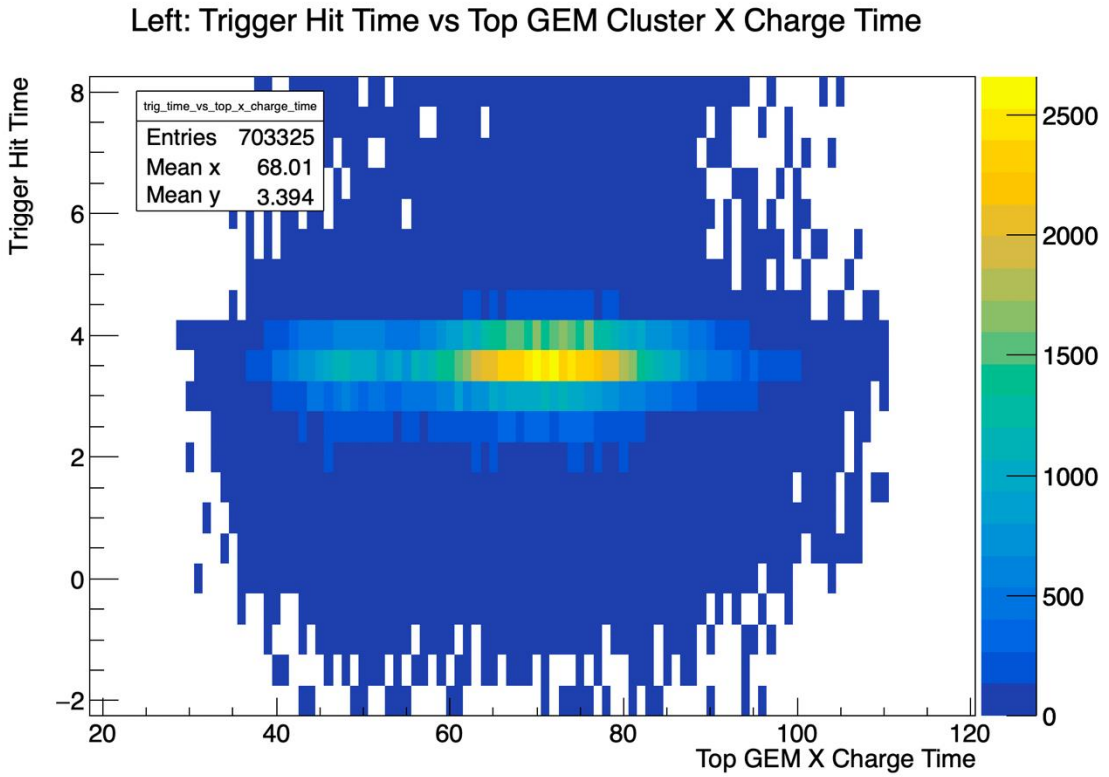
Single (one hit per detector per event) and On-Paddle (hit scintillators) tracks

Correlation Between Trigger and GEM Cluster Times

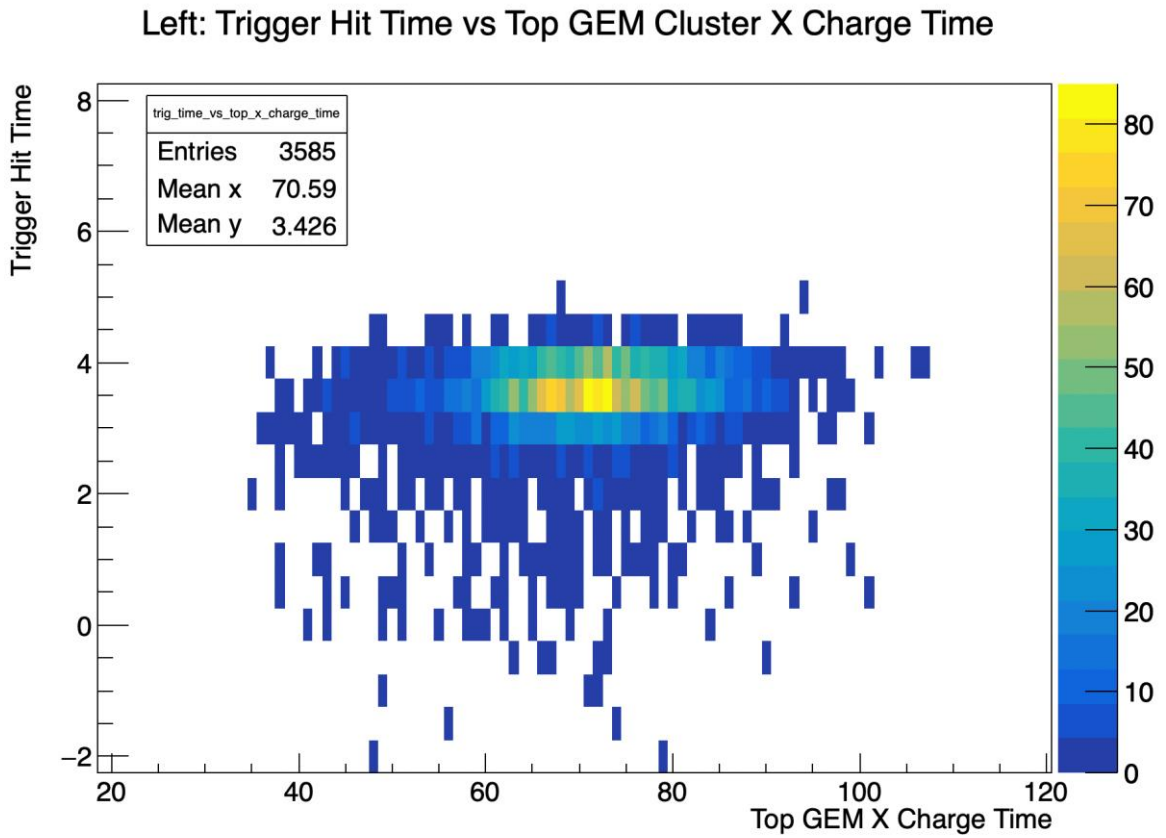
Charge Time

Trigger vs Left Top GEM X axis

Run 4010 - Elastic



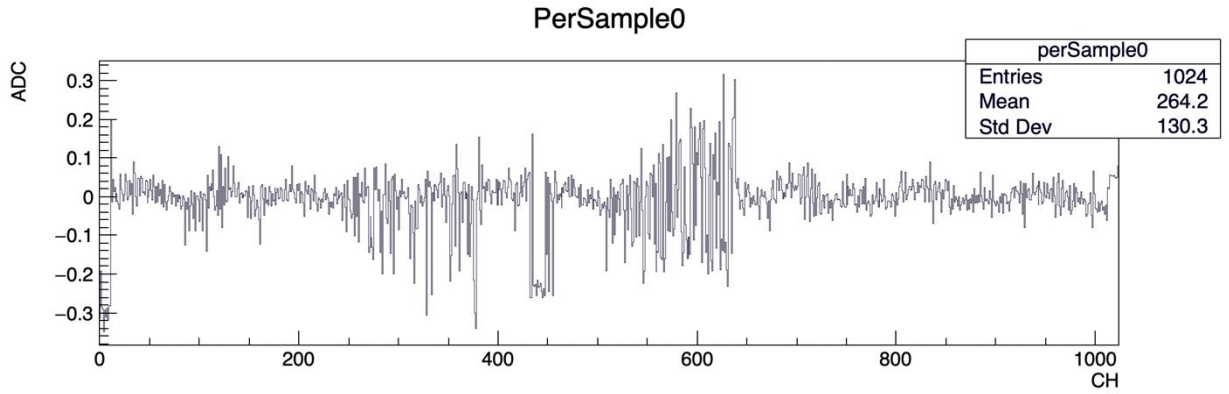
All tracks

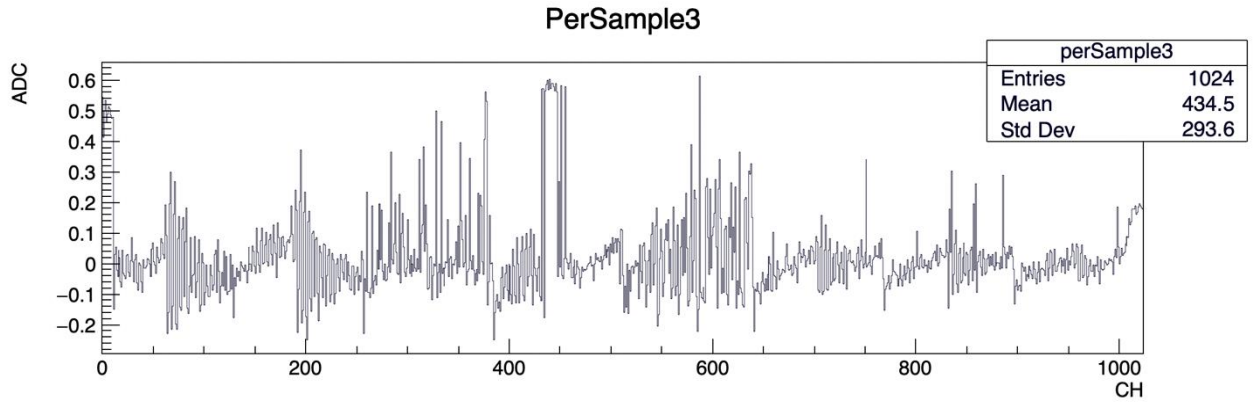


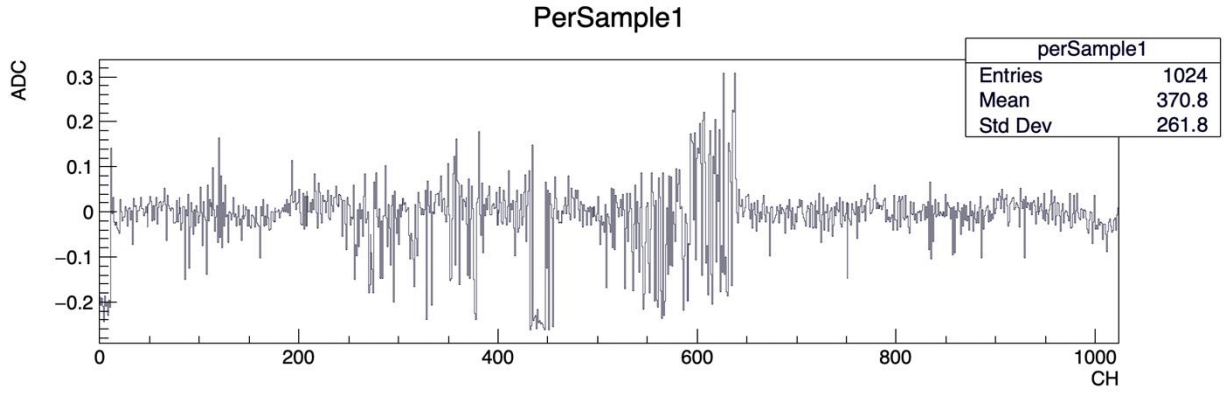
Single (one hit per detector per event) and On-Paddles (hits scintillators) tracks

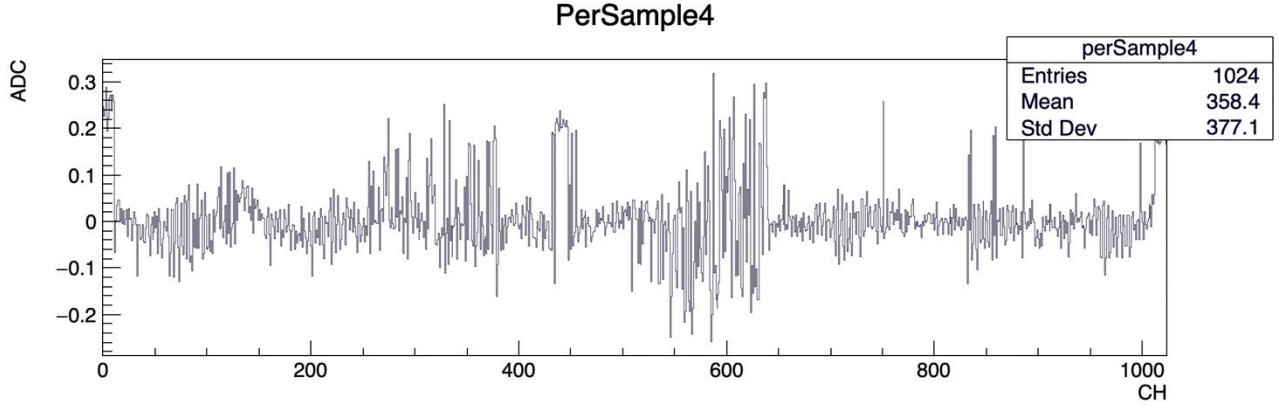
Review of cluster identification code in GEMini

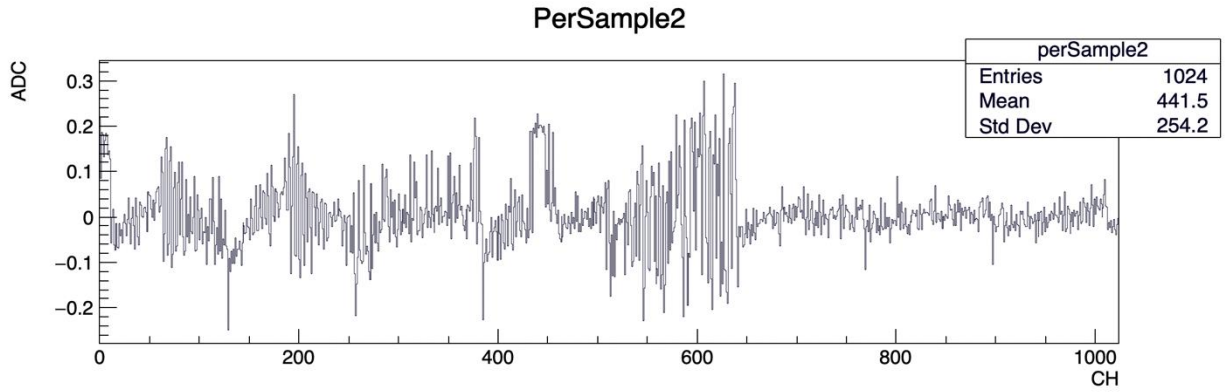
1. Pedestal-corrected data (Run 2679, Event 58, Left Top X)

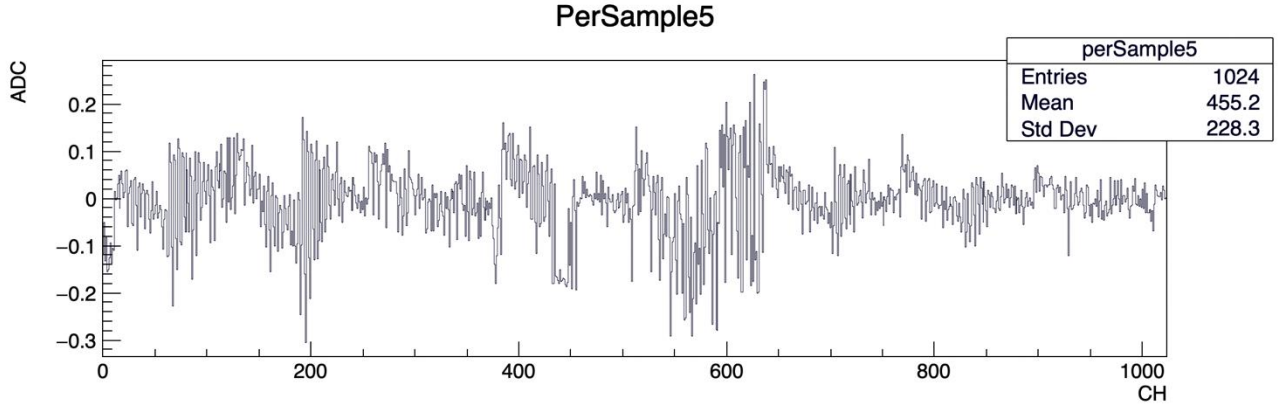




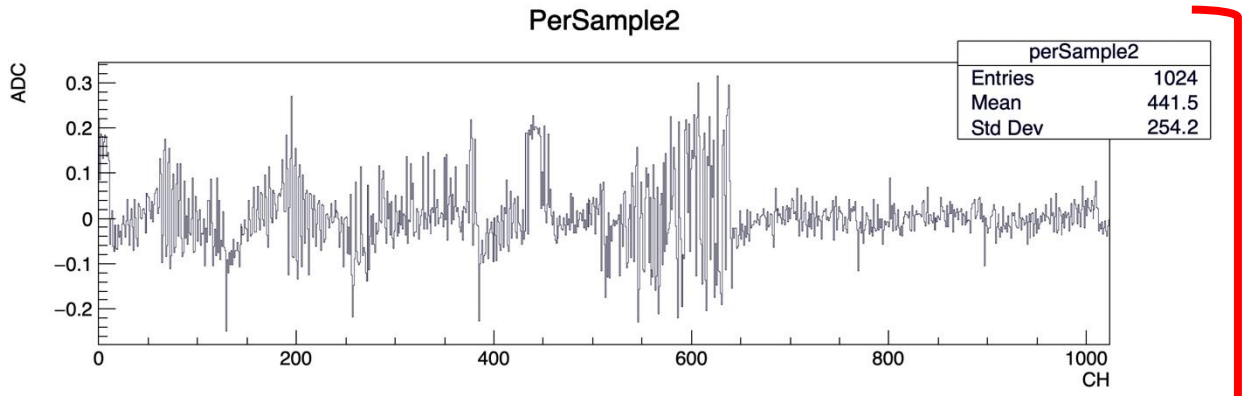


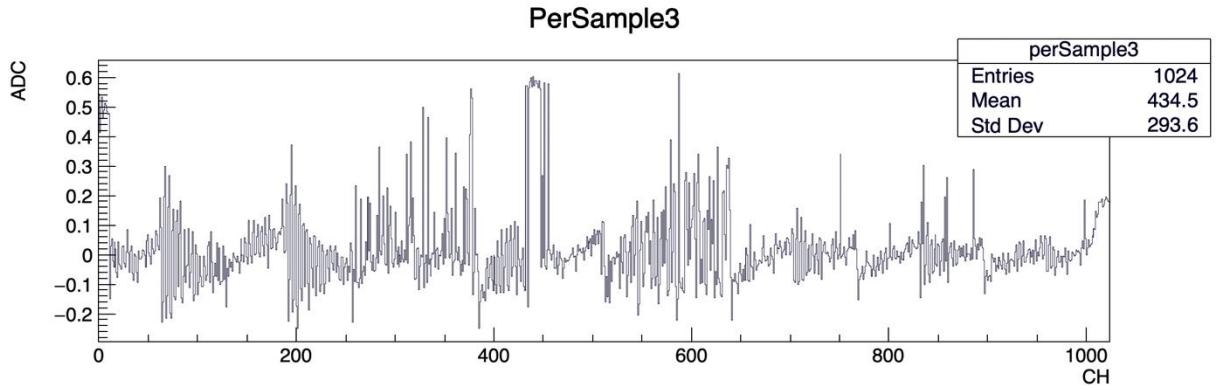


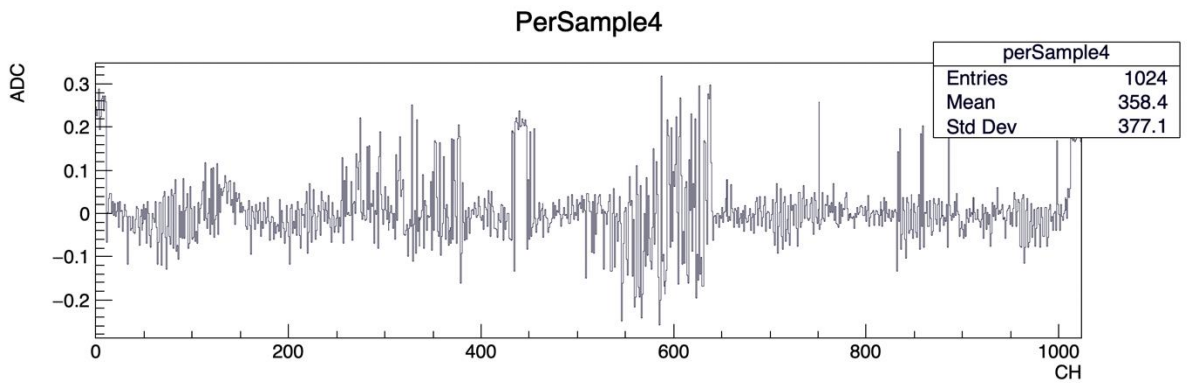


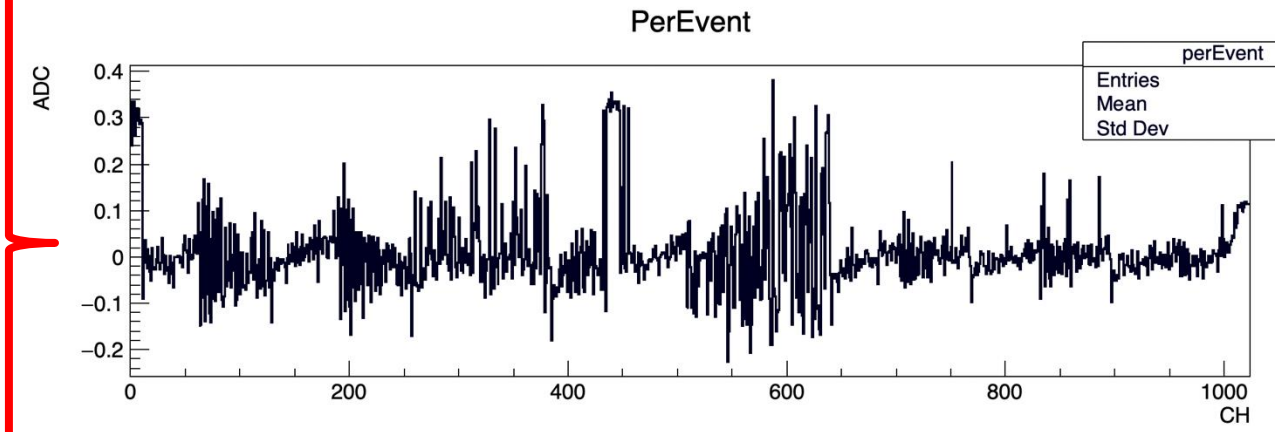


2. Time-Averaged Data (Run 2679, Event 58, Left Top X)









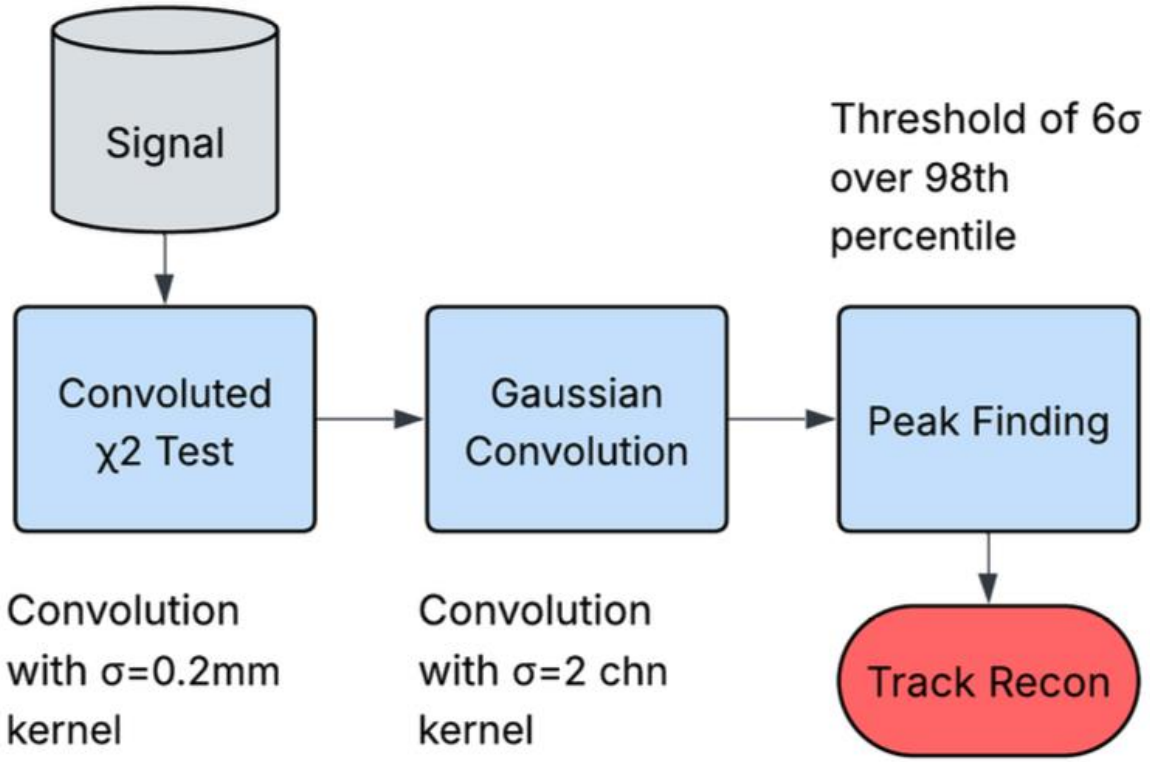


Figure 3.1: Pipline for GEM clustering

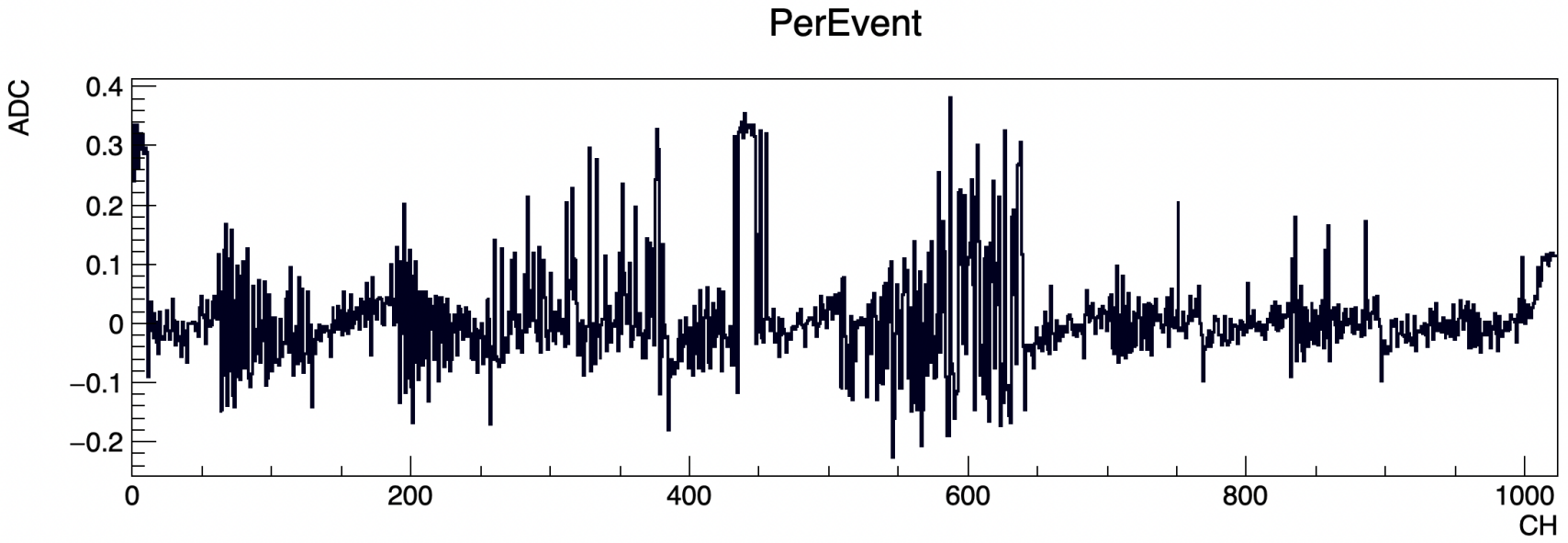
Credit: Andy Yang

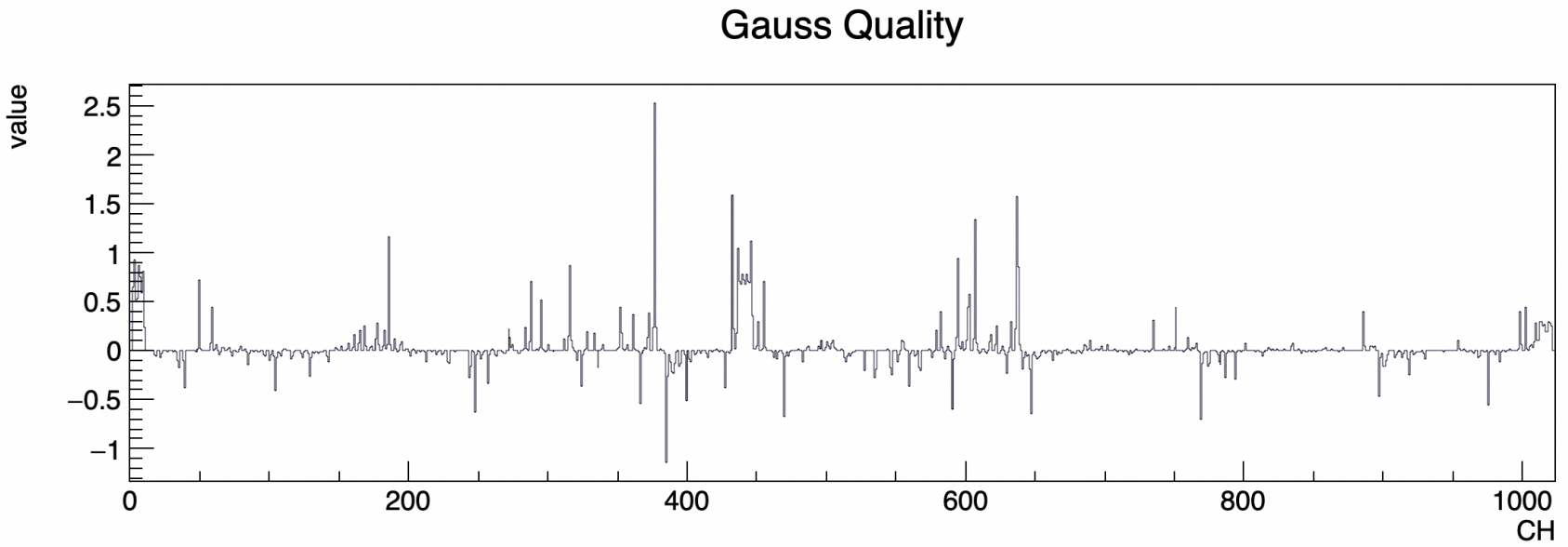
3. Convoluted χ^2 Test (Run 2679, Event 58, Left Top X)

Calculate “Gaussian quality” of each channel

How well window of five channels around each channel fits gaussian distribution with σ ≈ 1/2 channel width ≈ 0.2 mm

$$Q(i) = G(i)\,\frac{\chi^2_{\mathrm{NH}}(i) - \chi^2_{\mathrm{GH}}(i)}{\chi^2_{\mathrm{GH}}(i)}$$

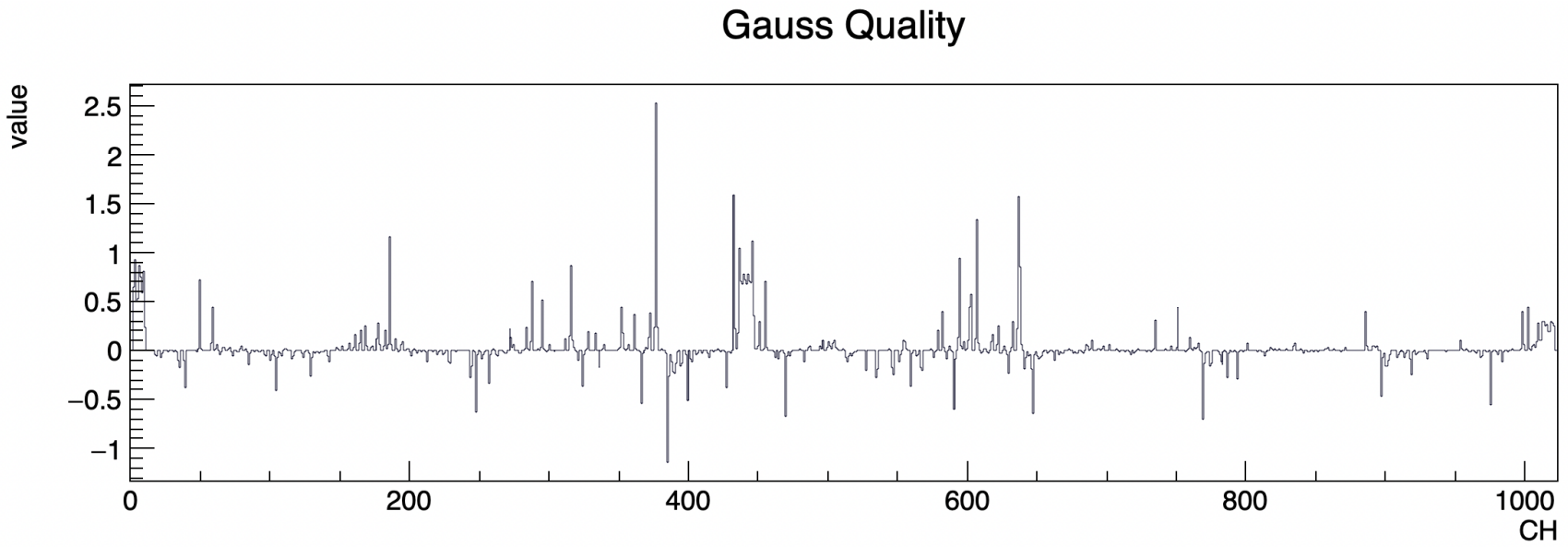


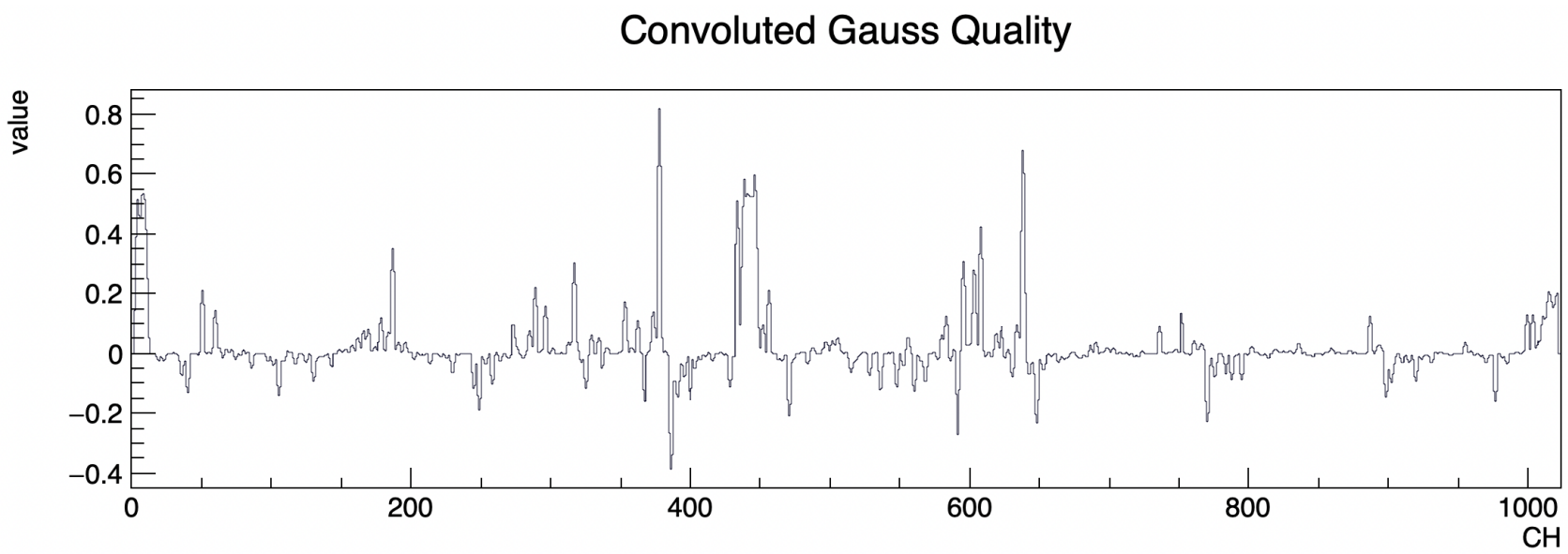


4. Gaussian Convolution (Run 2679, Event 58, Left Top X)

Only for X axis

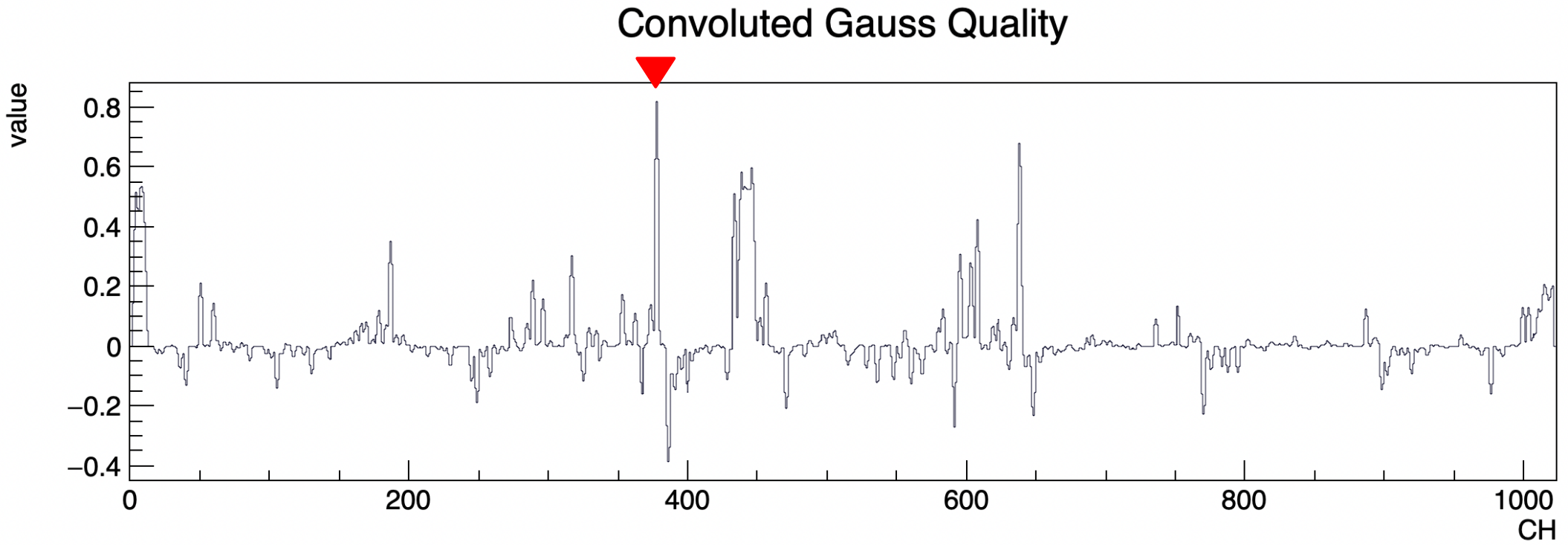
Convolute “Gaussian quality” with gaussian distribution of $\sigma \approx 2$ channels for window of five channels around each channel





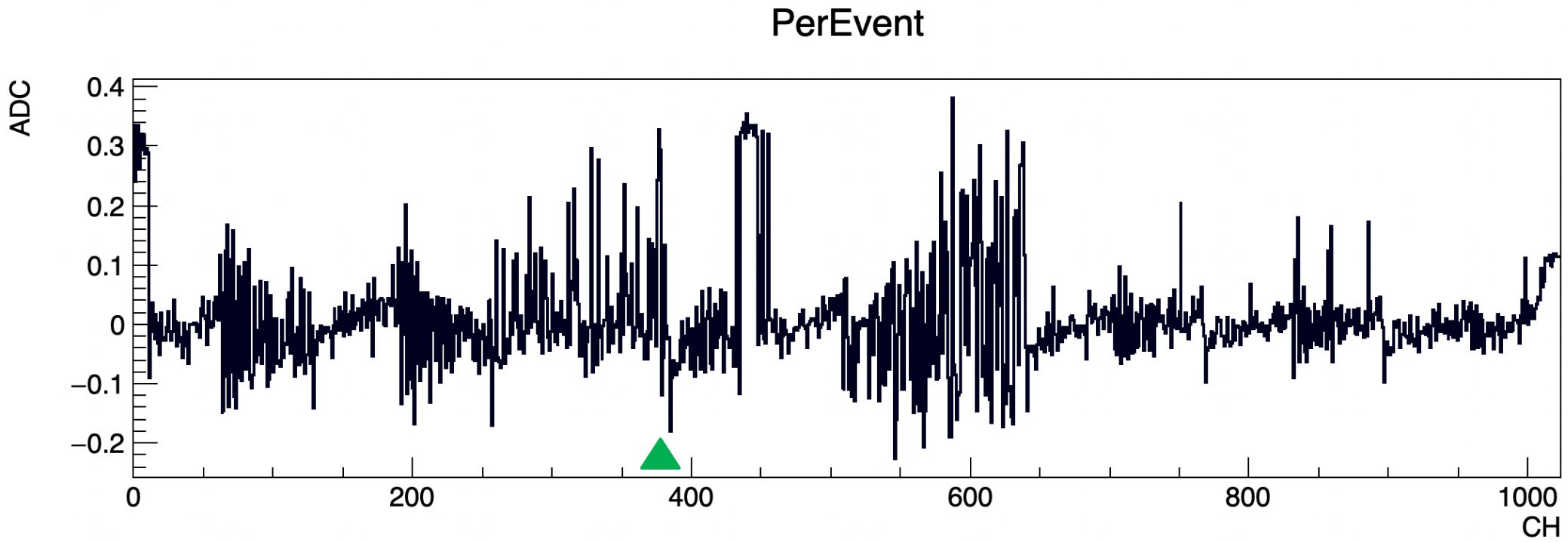
5. Peak finding – Quality (Run 2679, Event 58, Left Top X)

Peak threshold = median + 5 σ



5. Peak finding – ADC Value (Run 2679, Event 58, Left Top X)

Amplitude threshold = 5 σ

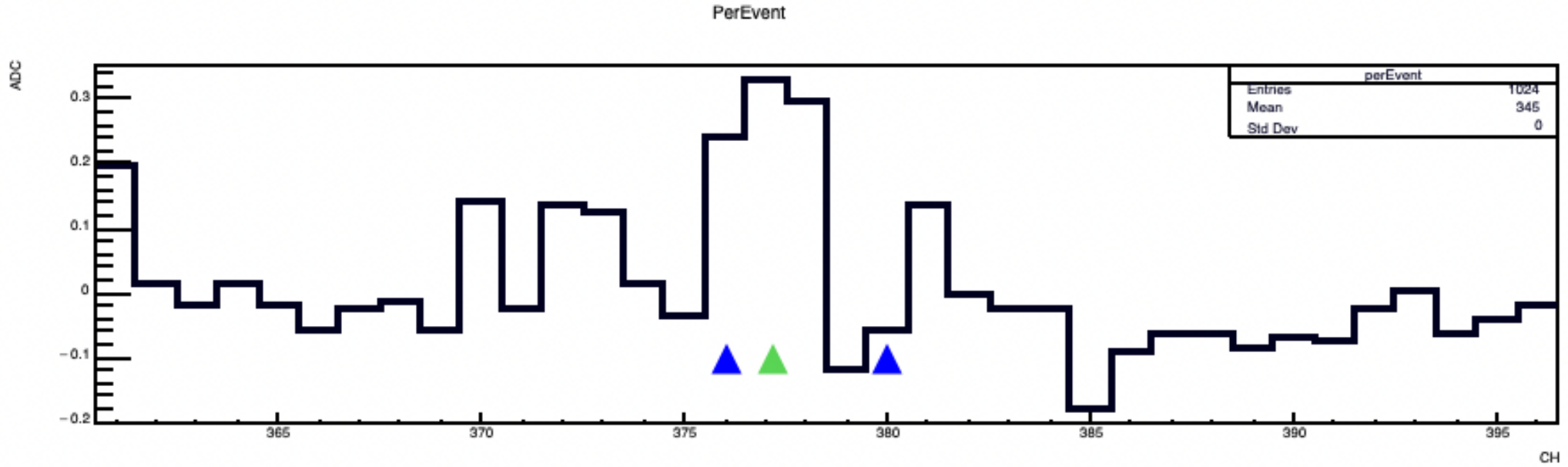


6. Position Reconstruction (Run 2679, Event 58, Left Top X)

”Charge Centroid Method”

Calculate mean position weighted by time-averaged channel values

- within range of channels where quality is over threshold



Range of channels where convoluted gauss quality is above threshold

Weighted mean position

N.B. I think there is an error in my code such that this range is off by one index

Ideas and potential next steps

- Think of ways to improve algorithm
  - Look at good clusters in good tracks?
  - Look at simulation?
- Improve identification of clusters in spatial domain
  - Optimize shape of Gaussian fits
  - Require threshold of signal above noise per APV instead of per axis?
- Analysis of clusters in time domain
  - Require threshold above time-domain noise (e.g. square wave noise)
  - Do some kind of fit?
  - Try time projection → calculate track angle?

Thank you